

Clasmo Diagnostics — LIMS

Complete Project Documentation

Laboratory Information Management System for Clasmo Diagnostics Pvt. Ltd.

Repository: https://github.com/Rudra19-ui/Clasmo_Diagnostics.git

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Stack: React (Vite) frontend + Django REST Framework backend

1. Project Overview

Clasmo Diagnostics is a multi-zone Laboratory Information Management System (LIMS). It covers patient search and registration, billing, barcode linking and scanning, sample workflow, clinical result entry, report verification, franchise wallets and pricing, administration masters, dashboards, reports, and home-collection device modules.

Architecture

Layer	Technology
Frontend	React 19, Vite 8, React Router 7, html5-qrcode
Backend	Django 6, Django REST Framework, Token authentication
Database	PostgreSQL 16 (Docker / production); SQLite possible locally
Media / PDF	Pillow, pypdf, WhiteNoise
Deploy	Docker, Gunicorn, Railway (railway.toml)
Legacy prototype	Static HTML/CSS/JS in repo root (index.html, css/, js/, device/)

Folder structure

```
CLASMO_DIAGNOSTICS/
├── backend/                Django API
│   ├── api/               models, views, serializers, permissions, tests
│   ├── clasmo_backend/    Django project settings
│   └── manage.py
├── frontend/              React SPA
│   └── src/
│       ├── components/    reusable UI
│       ├── pages/         route pages
│       ├── services/api.js HTTP client
│       ├── context/       Auth, Nav, ExtraSample
│       └── utils/         roles, nav, helpers
├── css/, js/, device/     legacy static prototype
├── docker-compose.yml
├── Dockerfile
└── docs/                  this documentation
```

2. Quick start

Backend

```
cd backend
python -m venv venv
source venv/bin/activate
pip install -r requirements.txt
python manage.py migrate
python manage.py seed_zones
python manage.py seed_test_roles
python manage.py seed_clinical_data
python manage.py runserver
# API: http://127.0.0.1:8000/api/
```

Frontend

```
cd frontend
npm install
npm run dev
# App: http://localhost:5173
```

Docker

```
docker-compose up
# API at http://localhost:8000
# Postgres: clasmo / clasmo / db clasmo
```

After production deploy, run `python manage.py migrate`.

3. Zones and demo logins

Operating zones: Nashik, Pune, Ratnagiri, Mumbai, Dhule. Zone admins only see their own zone. Patient bookings, doctors, and collection centers do not mix across zones.

Nashik demo accounts

Role	Username	Password
Admin	admin	admin123
User	user	user123
Technician	technician	tech123
Pathologist	pathologist	patho123
HR	hr	hr123
Receptionist	receptionist	reception123
Supreme	supreme	supreme123
Prime	prime	prime123

Sub-Franchise	sub	sub123
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Other zone admins

Zone	Username	Password
Pune	pune_admin	pune123
Ratnagiri	ratnagiri_admin	ratnagiri123
Mumbai	mumbai_admin	mumbai123
Dhule	dhule_admin	dhule123

Other-zone staff use {zone}_{role} usernames, for example mumbai_technician / mumbai_tech123.

4. User roles and access

Code	Label	Typical access
super_admin	Super Admin	Full system, all zones
admin	Admin	Zone admin, all operational modules
user	User	Front desk / general staff
technician	Technician	Sample workflow portal (same as reception) + clinical result entry APIs
receptionist	Receptionist	Sample workflow portal only
pathologist	Pathologist	Report verify, limited nav
hr	HR	Users, enquiries, administration
super_franchisee	Supreme	Top franchise: booking, pricing, wallets, Prime accounts
franchisee	Prime	Mid franchise: booking, Sub-Franchise pricing
sub_franchise	Sub-Franchise	Bottom franchise: booking and reports

Important role groups (frontend/src/utils/roles.js)

- **PATIENT_ENTRY_ROLES:** super_admin, admin, Supreme, Prime, Sub-Franchise
- **NO_PATIENT_ENTRY_ROLES:** pathologist, technician, receptionist, hr, user
- **RECEPTION_WORKFLOW_ROLES:** receptionist, technician (Sample Accession, Extra Sample, Outsource, Scan Log, etc.)
- **SAMPLE_SCAN_ROLES:** super_admin, admin, technician, pathologist
- **HOLD / REJECTION staff:** admin roles, hr, pathologist, technician, user (reception excluded from those sidebar items)
- **CLINICAL_ROLES:** super_admin, admin, technician, pathologist

Sidebar layouts

Receptionist and Technician use SAMPLE_WORKFLOW_PORTAL_NAV:

- Dashboard

- **SAMPLE WORKFLOW:** Sample Accession, Extra Sample, Out Sources, Out Received, Sample Rack, Sample Log, Scan Log
- **SETTINGS:** Change Password, Give Feedback, Help, Logout

Franchise roles use FRANCHISE_NAV (booking, reports, notifications, payments, pricing).

Admin / other staff use STANDARD_NAV (registration sections, scan, notifications, portfolio, administration).

Default LabRole permission keys

search, registration, test_result, administration, reports, device_request, dashboard, elab_pay, help, result_entry, report_preview, test_parameters, manage_users, manage_roles.

5. Database models

File: backend/api/models.py

Model	Purpose
Zone	Lab operating zones
User	Extended user: role, zone, franchise parent, display name
LoginLog	Login attempt audit
LabRole	Role permission JSON
TestCategory / Test / TestPackage	Test catalog, MRP, sample type, packages
ReportFormatAsset	Report format templates
Patient / PatientAddress	Patient master
Registration / RegistrationTest	Booking and ordered tests
RegistrationTestHold	Held tests by barcode
SampleRejection	Rejected samples
ExtraSample	Unlinked barcodes saved for follow-up
OutsourceTransfer	Samples sent to partner labs
PatientSampleBarcode	Barcode linked to patient / registration / sample type
BarcodeScanLog	Scan audit: barcode, zone, user, timestamp
PickupRequest / LabMessage	Home pickup and messages to lab
TestParameter / Report / ReportValue	Clinical parameters and results
InstrumentResultBatch	Analyzer ingest batches
MembershipType / Membership	Membership plans
CollectionCenter / CollectionCenterBoy	Centers and collection staff
Area / RateMaster	Geography and rate lists

DiscountReason / DiscountAuthority	Discount controls
WhatsAppMessageLog	WhatsApp delivery log
ExpenseType	Expense categories
Affiliation / SalesReference / Doctor	Referral masters
LabConfiguration	Lab profile, QR, next patient sequence
ServiceAreaPincode / LabActivity	Service pincodes and activities
JoinRequest / SelfPatientQuery	Public join and patient queries
FranchiseCommissionConfig	Commission rules
ZoneFranchiseRate / FranchisePricingOverride / FranchiseTestRate	Franchise pricing
FranchiseWallet / WalletTransaction / FranchiseLedgerEvent	Wallets, credits, ledger

6. Backend API endpoints

Base path: /api/. File: backend/api/urls.py.

Auth and health

Method	Endpoint	View	Function
GET	/health/	HealthView	Health check, optional seed
POST	/auth/login/	LoginView	Token login
POST	/auth/register/	RegisterView	Create account (allowed creators)
POST	/auth/logout/	LogoutView	Invalidate token
GET	/auth/me/	MeView	Current user
POST	/auth/change-password/	ChangePasswordView	Change password

Catalog

Method	Endpoint	View
GET	/tests/	TestListView
GET	/test-categories/	TestCategoryListView
GET	/test-packages/	TestPackageListView
GET	/report-formats/	ReportFormatListView

Registrations and billing

Method	Endpoint	View	Function
GET	/registrations/	RegistrationSearchView	Search bookings (franchise-scoped)

POST	/registrations/create/	RegistrationCreateView	New booking; wallet debit; next IDs
GET	/registrations/next-lab-code/	NextLabCodeView	Preview next lab code
GET	/registrations/next-patient-id/	NextPatientIdView	Preview next patient ID
GET	/registrations/<lab_code>/	RegistrationDetailView	Booking detail
PATCH	/registrations/<lab_code>/edit/	RegistrationEditView	Edit booking
POST	/registrations/<lab_code>/add-tests/	RegistrationAddTestsView	Add tests after booking
POST	/registrations/<lab_code>/cancel-tests/	RegistrationCancelTestsView	Cancel tests
POST	/registrations/<lab_code>/generate-mrp-bill/	RegistrationGenerateMrpBillView	MRP bill
POST	/registrations/<lab_code>/clinical-pdf/	RegistrationClinicalPdfView	Upload clinical PDF
GET	/registrations/worksheet/	WorksheetView	Printable worksheet
GET	/registrations/workflow/	WorkFlowHistoryView	Status history
GET	/registrations/barcode/	BarcodeView	Barcode print payload
GET/PATCH	/registrations/bill-receipt/	BillReceiptView	Bill receipt
GET/POST	/registrations/notifications/	NotificationView	SMS / WhatsApp
POST	/registrations/notifications/bulk/	BulkNotificationView	Bulk release SMS

Sample workflow

Method	Endpoint	View	Function
GET	/sample-accession/	SampleAccessionListView	All zone bookings for accession
GET	/scan-logs/	ScanLogListView	Scan audit grouped by zone
GET/POST	/holds/	RegistrationTestHoldListCreateView	List/create holds
POST	/holds/<id>/release/	RegistrationTestHoldReleaseView	Release hold
GET/POST	/rejections/	SampleRejectionListCreateView	List/create rejections
POST	/rejections/<id>/resolve/	SampleRejectionResolveView	Resolve rejection
GET/POST	/extra-samples/	ExtraSampleListCreateView	Save unlinked barcodes
POST	/extra-samples/<id>/remove/	ExtraSampleRemoveView	Remove extra sample
GET/POST	/outsource-transfers/	OutsourceTransferListCreateView	Send outsource
POST	/outsource-transfers/receive/	OutsourceTransferReceiveView	Receive by barcode/lab
POST	/outsource-transfers/<id>/receive/	OutsourceTransferReceiveByIdView	Receive by id
POST	/outsource-transfers/<id>/upload-report/	OutsourceTransferUploadReportView	Upload partner report

Barcodes

Method	Endpoint	View
GET	/patient-barcodes/	PatientBarcodeListView
GET	/patient-barcodes/lookup/	PatientBarcodeLookupView
GET	/patient-barcodes/scan/	PatientSampleScanView (also writes BarcodeScanLog)
POST	/patient-barcodes/link/	PatientBarcodeLinkView
GET/DELETE	/patient-barcodes/<id>/	PatientBarcodeDetailView

Clinical and instruments

Method	Endpoint	View
GET/POST	/test-parameters/	TestParameterListCreateView
GET/PUT/DELETE	/test-parameters/<id>/	TestParameterDetailView
GET/POST	/reports/<registration_id>/	ReportDetailView
PATCH	/reports/<registration_id>/verify/	ReportVerifyView
POST	/public/patient-report/	PublicPatientReportView
POST	/instrument/results/	InstrumentResultsIngestView
GET	/instrument/patient-report/	PatientReportByBarcodeView

Franchise wallet and pricing

Method	Endpoint	View
GET	/franchise/ledger/	FranchiseLedgerView
GET	/franchise/sample-usage/	FranchiseSampleUsageView
GET	/wallets/me/	MyWalletView
GET	/wallets/	WalletListView
GET	/wallets/<user_id>/	WalletDetailView
GET	/wallets/transactions/	WalletTransactionListView
GET/PUT	/wallets/commission-config/	CommissionConfigView
GET/PUT	/wallets/zone-franchise-rates/	ZoneFranchiseRateListView
GET	/wallets/zone-franchise-rates/me/	MyZoneFranchiseRateView
GET/PUT	/wallets/pricing-overrides/me/	MyPricingOverrideView
POST	/wallets/top-up/	WalletTopUpView
POST	/wallets/adjust/	WalletAdjustmentView
POST	/wallets/online-top-up/	WalletSelfTopUpView
GET/PUT	/wallets/franchise-test-rates/	FranchiseBulkPricingView

POST	/wallets/franchise-test-rates/transfer/	FranchiseTransferPricingView
POST	/wallets/demo-transaction/	DemoWalletTransactionView

Dashboard, search, device

Method	Endpoint	View
GET	/dashboard/summary/	DashboardSummaryView
GET	/reports/summary/	ReportSummaryView
GET	/search/global/	GlobalSearchView
GET/POST	/pickup-requests/	PickupRequestListCreateView
GET/POST	/messages/	LabMessageListCreateView

Administration masters

CRUD or list endpoints exist for: users, user role update, zones, roles, membership types/memberships, collection centers, areas, rate masters, collection center boys, discount reasons/authorities, WhatsApp logs, expense types, doctors, affiliations, sales references, patients, lab configuration + QR upload, service area pincodes, activities, join requests, self-patient queries.

7. Backend services and functions

Permissions (clinical_permissions.py)

- CanAccessPatientEntry, CanAccessPricingWallet, CanManageFranchiseTestRates
- CanAccessHolds, CanPlaceHolds, CanAccessRejections, CanRejectSamples
- CanAccessSampleAccession (receptionist + technician)
- CanAccessExtraSamples, CanAccessScanLog (reception, technician, admin, super_admin)
- CanAccessOutsource
- IsAdmin, IsClinicalStaff, IsTechnicianOrAdmin, IsPathologistOrAdmin, TestParameterPermission

Franchise / zone scoping (franchise_scope.py)

- is_franchise_user, user_zone_id, user_has_global_data_access
- franchise_descendant_ids, visible_creator_ids
- scope_registrations_for_user — hierarchy-scoped bookings
- scope_zone_registrations_for_accession — all bookings in the user's zone
- scope_patients_for_user, scope_reports_for_user, scope_barcode_for_user
- scope_created_by_for_user, scope_zone_queryset, scope_users_for_user
- user_can_access_registration, get_registration_for_user
- scope_wallets_for_user, scope_wallet_transactions_for_user, user_can_access_wallet

Barcode service (barcode_service.py)

- normalize_barcode, validate_barcode_pair, link_sample_barcodes
- resolve_patient_for_link, lookup_patient_by_barcode, barcode_lookup_candidates
- filter_registrations_by_barcode, sample_group_for_test, test_matches_sample_type
- build_sample_scan_payload, scan_sample_by_barcode
- mark_registration_sample_scanned, mark_sample_barcode_scanned
- record_barcode_scan_log — writes BarcodeScanLog with username

Wallet service (wallet_service.py)

- get_or_create_wallet, hierarchy_beneficiaries, commission_beneficiaries
- distribute_registration_commissions, settle_registration_booking
- settle_registration_test_addition, debit_registration_charge
- admin_adjust_wallet, admin_set_wallet_balance, admin_top_up_wallet
- franchise_online_top_up, can_release_report, assert_can_release_report
- apply_ledger_entry, create_demo_transactions

Main API views (views.py) — class list

HealthView, LoginView, RegisterView, LogoutView, MeView, ChangePasswordView, TestListView, TestCategoryListView, TestPackageListView, ReportFormatListView, RegistrationSearchView, SampleAccessionListView, ScanLogListView, WorksheetView, WorkFlowHistoryView, BarcodeView, NotificationView, BulkNotificationView, BillReceiptView, RegistrationDetailView, RegistrationClinicalPdfView, RegistrationEditView, RegistrationAddTestsView, RegistrationCancelTestsView, RegistrationTestHoldListCreateView, RegistrationTestHoldReleaseView, SampleRejectionListCreateView, SampleRejectionResolveView, ExtraSampleListCreateView, ExtraSampleRemoveView, OutsourceTransferListCreateView, OutsourceTransferReceiveView, OutsourceTransferReceiveByIldView, OutsourceTransferUploadReportView, RegistrationCreateView, RegistrationGenerateMrpBillView, FranchiseLedgerView, FranchiseSampleUsageView, NextLabCodeView, NextPatientIldView, PickupRequestListCreateView, LabMessageListCreateView, DashboardSummaryView, ReportSummaryView, UserListView, ZoneListView, UserRoleUpdateView, RoleListView, RoleDetailView, MembershipTypeListView, MembershipListCreateView, CollectionCenterListCreateView, CollectionCenterDetailView, AreaListView, RateMasterListView, CollectionCenterBoyListCreateView, CollectionCenterBoyDetailView, DiscountReasonListCreateView, DiscountReasonDetailView, DiscountAuthorityListCreateView, DiscountAuthorityDetailView, WhatsAppLogListView, ExpenseTypeListCreateView, ExpenseTypeDetailView, DoctorListCreateView, DoctorDetailView, AffiliationListView, SalesReferenceListView, PatientMasterListCreateView, PatientMasterDetailView, LabConfigurationView, LabQrCodeUploadView, ServiceAreaPincodeListCreateView, ServiceAreaPincodeDetailView, LabActivityListCreateView, LabActivityDetailView, JoinRequestListCreateView, JoinRequestDetailView, SelfPatientQueryListCreateView, SelfPatientQueryDetailView, GlobalSearchView, PatientBarcodeListView, PatientBarcodeLookupView, PatientSampleScanView, PatientBarcodeLinkView, PatientBarcodeDetailView.

Clinical views

TestParameterListCreateView, TestParameterDetailView, ReportDetailView, PublicPatientReportView, ReportVerifyView.

Wallet views

MyWalletView, WalletListView, WalletDetailView, WalletTransactionListView, CommissionConfigView, ZoneFranchiseRateListView, MyZoneFranchiseRateView, MyPricingOverrideView, WalletTopUpView, WalletAdjustmentView, WalletSelfTopUpView, DemoWalletTransactionView, FranchiseBulkPricingView, FranchiseTransferPricingView.

Instrument views

InstrumentResultsIngestView, PatientReportByBarcodeView.

8. Frontend pages and routes

Public

Route	Page
/	Landing (Join with Clasmo, departments, public patient report)
/login	Login
/test-quorum	TestQuorum

Core authenticated

Route	Page	Roles
/dashboard	Dashboard	All logged-in
/search	Search	Patient entry
/open	OpenBarcode	Logged-in
/barcode-print	BarcodePrinting	Logged-in
/barcode-link	BarcodeLink	Patient entry
/sample-scan	SampleScan	Admin, technician, pathologist
/hold-tests	HoldTests	Hold staff
/sample-rejection	SampleRejection	Rejection staff
/bill-receipt	BillReceipt	Patient entry
/test-result / test-result-entry	TestResult / TestResultEntry	Patient entry / logged-in
/entry/new, /entry/list, /entry/test-addition	Registration, EditEntry, TestAddition	Patient entry
/reports-section/all search detail/:labCode	AllReports, SearchReports, ReportDetail	Patient entry
/reports	Reports	Patient entry
/portfolio/*	TestList, TestProfile, SampleReport	Test portfolio roles

/user-signup	UserSignUp	Creators
/enquire-box	EnquireBox	Admin
/self-patient-query	SelfPatientQuery	Logged-in
/give-feedback	GiveFeedback	Logged-in
/help, /elab-pay	Help, ElabPay	Logged-in

Reception / technician sample workflow

Route	Page	Status
/reception/sample-accession	SampleAccession	Live zone-wide list
/reception/extra-sample	Scan (title Extra Sample)	Live scan + save extra sample
/reception/out-sources	OutSources	Live send outsource
/reception/out-received	OutReceived	Live receive + upload report
/reception/scan-log	ScanLog	Live audit log with username
/reception/sample-rack	ReceptionModule stub	Placeholder
/reception/sample-log	ReceptionModule stub	Placeholder
/notifications/scan	Scan	Staff/franchise scan UI

Clinical

Route	Page	Roles
/clinical/test-parameters	TestParameterMaster	Admin
/clinical/result-entry	ResultEntry	Admin, technician
/clinical/report-preview	ReportPreview	Logged-in

Franchise portal

Route	Page
/franchise/analytics	Analytics (Supreme, Prime)
/franchise/manage-booking/new list test-addition	Registration, EditEntry, TestAddition
/franchise/manage-reports/all search detail/:labCode	Reports
/franchise/find-barcode	FindBarcode
/franchise/clinical-history	ClinicalHistory
/franchise/test-cancellation	TestCancellation
/franchise/hold / rejection	FranchiseHold, FranchiseRejection
/franchise/make-bill / billing-list / track-ledger	MakeBill, BillingList, TrackLedger
/franchise/pricing-credits	FranchisePricingCredits

/franchise/online-payment	OnlinePayment
/franchise/franchisee-pricing and transfer	Supreme bulk/transfer pricing for Prime
/franchise/sub-franchisee-pricing and transfer	Prime bulk/transfer pricing for Sub
/franchise/scan	Scan
/franchise/payment-history, my-staff, sub-franchisee, update-profile	FranchiseStub placeholders

Admin hub (/administration and /admin/*)

Live pages: Change Password, Role Management, Membership, Collection Center Boy, Discount Reason/Authority, WhatsApp Logger, Expense Type, Collection Center Management, Doctor Management, Patient Management, Lab Configuration, Services In Area, Create Activity, Activities, Franchise Credits, List Franchisee, Add Franchisee, Franchise Bulk Pricing, Transfer Price.

Stub modules via AdminModule: User Lock, User Management, CRM, QC Report, ICMR Batches, Download offline data, CheckList, Machine Mapping, Area Master, Sales Reference Master, Lab Accounting, Billing, Rate Master, Test Details/Category/Group/Profile/Unit, Special Offers, Notes, Machine Interface, Popular Test, Complete Test List, OutSource Lab, External Test Mapping.

Device

Route	Page
/device/pickup-request	PickupRequest (live)
/device/message-to-lab	MessageToLab (live)
/device/test-result-batch	TestResultBatch
/device/patient-appointment, scheduler, trip-management, batch-upload	DeviceStub

9. Frontend API client (api.js)

Token stored as `clasmo_token`. Base URL: `VITE_API_URL` or `/api`.

Auth: login, register, logout, me, changePassword.

Catalog: getTests, getTestCategories, getTestPackages, getReportFormats.

Registrations: searchRegistrations, listSampleAccession, listScanLogs, getWorksheet, getWorkflowHistory, getBarcodeData, getBillReceipt, getMultipleBillReceipts, saveBillReceipt, getNotificationPrefill, sendNotification, sendBulkNotification, createRegistration, getNextLabCode, getNextPatientId, getRegistration, uploadRegistrationClinicalPdf, updateRegistration, addRegistrationTests, cancelRegistrationTests, generateMrpBill.

Workflow: listHolds, createHolds, releaseHold, listRejections, createRejection, resolveRejection, listExtraSamples, createExtraSample, removeExtraSample, listOutsourceTransfers, createOutsourceTransfer, receiveOutsourceTransfer, receiveOutsourceTransferById, uploadOutsourceReport.

Barcodes: getPatientBarcodes, lookupPatientBarcode, scanSampleBarcode, linkPatientBarcodes.

Clinical: getTestParameters, create/update/delete TestParameter, getReport, submitReport, verifyReport, ingestInstrumentResults, getPatientReportByBarcode.

Wallets: getMyWallet, getWallets, getWalletTransactions, zone franchise rates, pricing overrides, topUpWallet, adjustWallet, onlineTopUpWallet, getFranchiseTestRates, saveFranchiseTestRates, transferFranchiseTestRates.

Admin CRUD: users, roles, memberships, collection centers, doctors, patients, lab config, activities, join requests, self-patient queries, discounts, WhatsApp logs, expense types, service pincodes, areas, rate masters.

Other: getDashboardSummary, getReportSummary, pickup requests, messages, globalSearch, getPublicPatientReport, submitJoinRequest.

10. Frontend components

Component	Role
Layout / FranchiseSidebar / AdminSidebar	App chrome and nav
ProtectedRoute	Auth + allowedRoles
QrCameraScanner	Camera barcode/QR scan
BarcodeLinkForm	Link barcodes to patient
ExtraSampleNoDataModal / Panel	No-data scan → Extra Sample
SampleScanResultPanel	Scan result display
ReportSectionTable	Booking table (accession/reports)
WorkFlowHistoryModal	Registration workflow events
SendSMSModal / BulkReleaseSMSModal	Notifications
TestBillModal / TestDualListPicker	Billing and test pickers
WalletBalanceBadge	Header wallet
FranchiseBulkPricingPanel / TransferPricingPanel	Pricing UI
DashboardCharts / PatientStatusBadge	Dashboard
DatePicker, GenderRadioGroup, DataTable, FilterPanel, StatusTabs	Forms and tables
JoinWithClasmo, DepartmentGrid, LandingBrandTitle	Public landing
EnquireBoxPanel, SelfPatientQueryPanel, FeedbackPanel	Queries and feedback

Context

- AuthContext — login state, user, token
- NavContext — mobile drawer and sidebar scroll
- ExtraSampleContext — saved extra-sample barcodes

11. Feature modules (what they do)

Registration / New Entry

Creates patient + registration + tests. Uses LabConfiguration.next_patient_sequence (no full-table MAX scan). Returns next_lab_code and next_patient_id. Debits franchise wallet and distributes cascade commissions. Clinical PDF upload is non-blocking after success.

Sample Accession

Reception and technician see all bookings in their zone regardless of creator role. Uses ReportSectionTable format (Booking ID, Patient, Barcodes, Ref By, Tests, Status).

Scan / Extra Sample

Scan or type barcode. If found, show patient/sample. If not found, option to save as Extra Sample. Extra samples are zone-scoped.

Scan Log

Audit of successful scans from PatientSampleScanView. Shows barcode, patient, lab code, sample type, scanned-by username, time. Reception/technician see own zone; global admins see all zones in separate sections. Historical scans before this feature are not backfilled.

Out Sources / Out Received

Send selected tests to partner labs; receive them back; optional report file upload. Row coloring indicates status.

Hold and Rejection

Staff place holds or reject samples by barcode. Franchise users see holds/rejections on their hierarchy entries.

Clinical

Parameter master (admin). Result entry (technician/admin). Pathologist verifies. Instrument ingest can post analyzer results. Public patient report lookup exists.

Franchise money

Per-user wallets. Booking charges debit the creator. Commissions credit Supreme → Prime → Sub cascade. Admin can top up/adjust. Online top-up for franchise. Bulk per-test rates and transfer pricing from parent to child franchisees. Report release can be blocked if wallet rules fail.

Zone isolation

Most operational data is filtered by user.zone. Super admin / selected global roles can see all zones.

12. Management commands

Command	Purpose
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seed_data	Initial demo catalog/data
seed_zones	Create five zones and zone admins
seed_test_roles	Create role accounts per zone
seed_clinical_data	Clinical parameters
ensure_trial_users	Ensure trial users exist
import_por_catalog	Import POR test catalog
import_ratnagiri_rates	Import Ratnagiri rates

13. Automated tests

File	Coverage
test_sample_accession.py	Zone-wide accession; technician access
test_scan_log.py	Scan log username, zone isolation, technician list
test_extra_sample.py	Extra sample create/list
test_outsource_transfer.py	Outsource send/receive
test_patient_entry_permissions.py	Who can book
test_franchise_scope.py	Hierarchy scoping
test_zone_isolation.py	Zone data isolation
test_wallet_credits.py / test_pricing_wallet.py	Wallet and pricing
test_cascade_pricing.py / test_franchise_bulk_pricing.py	Cascade and bulk rates
test_registration_add_tests.py / test_registration_edit.py	Add tests and edit
test_sample_barcode_filter.py	Barcode search
test_instrument_ingest.py	Analyzer ingest

14. Migrations of note

Migration	Change
0034	Franchise wallet credit
0038	Zone structure
0041	Franchise test rate
0044	Sample rejection
0046 / 0047	Outsource transfer + test IDs
0048	LabConfiguration.next_patient_sequence
0049	Extra Sample
0050	BarcodeScanLog

15. Environment and deploy

- Backend env: DATABASE_URL, SECRET_KEY, DEBUG, ALLOWED_HOSTS, RUN_SEED, PORT
- Frontend env: VITE_API_URL
- Docker Compose: postgres:16-alpine + web build, RUN_SEED=true
- entrypoint.sh runs migrations (and optional seed) then gunicorn
- Auth header: Authorization: Token <key>

Production checklist: migrate through 0050; keep zone isolation; Scan Log only records scans after that feature went live; Sample Rack and Sample Log pages are still placeholders.

16. Tech dependencies

Python: django 6.x, djangorestframework 3.17, django-cors-headers, pillow, pypdf, gunicorn, whitenoise, psycopg2-binary, dj-database-url, python-dotenv.

Node: react 19, react-dom, react-router-dom 7, html5-qrcode, vite 8.

This document describes the Clasmo Diagnostics LIMS as of 4 September 2026. Source of truth is the GitHub repository. HTML source: docs/CLASMO_PROJECT_DOCUMENTATION.html